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B.Sc. in Applied Sciences

MAT 2306 – Operational Research I
Tutorial No: 1

1. Choose the correct answer for following questions:

- I. Operations research is the application of ----- methods to arrive at the optimal Solutions to the problems.
 - A. economical
 - B. scientific
 - C. a and b both
 - D. artistic
- II. In operations research, the -----are prepared for situations.
 - A. mathematical models
 - B. physical models diagrammatic
 - C. diagrammatic models
- III. Operations management can be defined as the application of -----
-----to a problem within a system to yield the optimal solution.
 - A. Suitable manpower
 - B. mathematical techniques, models, and tools
 - C. Financial operations
- IV. Operations research is based upon collected information, knowledge and advanced study of various factors impacting a particular operation. This leads to more informed -----.
 - A. Management processes
 - B. Decision making
 - C. Procedures
- V. OR can evaluate only the effects of -----
 - A. Personnel factors.
 - B. Financial factors
 - C. Numeric and quantifiable factors.

- VI. The objective function and constraints are functions of two types of variables, ----- variables and ----- variables.
- A. Positive and negative
 - B. Controllable and uncontrollable
 - C. Strong and weak
 - D. None of the above
- VII. Who defined OR as scientific method of providing executives departments with a quantitative basis for decisions regarding the operations under their control?
- A. Morse and Kimball (1946)
 - B. P.M.S. Blackett (1948)
 - C. E.L. Arnoff and M.J. Netzorg
 - D. None of the above
- VIII. OR has a characteristic that it is done by a team of
- A. Scientists
 - B. Mathematicians
 - C. Academics
 - D. All of the above
- IX. What have been constructed from OR problems a method for solving the models that are available in many cases?
- A. Scientific Models
 - B. Algorithms
 - C. Mathematical Models
 - D. None of the above
- X. Which technique is used in finding a solution for optimizing a given objective, such as profit maximization or cost reduction under certain constraints?
- A. Quailing Theory
 - B. Waiting Line
 - C. Both A and B
 - D. Linear Programming
- XI. What is the objective function in linear programming problems?
- A. A constraint for available resource
 - B. An objective for research and development of a company
 - C. A linear function in an optimization problem
 - D. A set of non-negativity conditions

- XII. Which statement characterizes standard form of a linear programming problem?
- Constraints are given by inequalities of any type
 - Constraints are given by a set of linear equations
 - Constraints are given only by inequalities of $\geq$ type
 - Constraints are given only by inequalities of $\leq$ type
- XIII. In graphical representation the bounded region is known as ----- region.
- Solution
 - basic solution
 - feasible solution
 - optimal
- XIV. In Degenerate solution value of objective function -----
- increases infinitely
 - basic variables are nonzero
 - decreases infinitely
 - One or more basic variables are zero
- XV. In LPP the condition to be satisfied is
- Constraints have to be linear
 - Objective function has to be linear
 - none of the above
 - both a and b
2. A company manufactures two products X and Y, which require, the following resources. The resources are the capacities machine M_1 , M_2 , and M_3 . The available capacities are 50, 25, and 15 hours respectively in the planning period. Product X requires 1 hour of machine M_2 and 1 hour of machine M_3 . Product Y requires 2 hours of machine M_1 , 2 hours of machine M_2 and 1 hour of machine M_3 . The profit contribution of products X and Y are Rs.5/-and Rs.4/- respectively.

The contents of the statement of the problem can be summarized as follows

Machines	Products		Availability in hours
	X	Y	
M1	0	2	50
M2	1	2	25
M3	1	1	15
Profit in Rs. Per unit	5	4	

In the above problem, Products X and Y are competing candidates or variables. Formulate a mathematical model for the problem.

3. A retail store stocks two types of shirts A and B. These are packed in attractive cardboard boxes. During a week the store can sell a maximum of 400 shirts of type A and a maximum of 300 shirts of type B. The storage capacity, however, is limited to a maximum of 600 of both types combined. Type A shirt fetches a profit of Rs. 2/- per unit and type B a profit of Rs. 5/- per unit. Formulate a mathematical model for how should stock per week to maximize the total profit.
4. A ship has three cargo holds, forward, aft and center. The capacity limits are:
 Forward 2000 tons, 100,000 cubic meters
 Center 3000 tons, 135,000 cubic meters
 Aft 1500 tons, 30,000 cubic meters.

The following cargoes are offered, the ship owners may accept all or any part of each commodity:

Commodity	Amount in tons.	Volume/ton in cubic meters	Profit per ton in Rs.
A	6000	60	60
B	4000	50	80
C	2000	25	50

In order to preserve the trim of the ship the weight in each hold must be proportional to the capacity in tons. Formulate this as linear programming problem for How should the cargo be distributed so as to maximize profit.

5. A patient consults a doctor to check up his ill health. Doctor examines him and advises him that he is having deficiency of two vitamins, vitamin A and vitamin D. Doctor advises him to consume vitamin A and D regularly for a period of time so that he can regain his health. Doctor prescribes tonic X and tonic Y, which are having vitamin A, and D in certain proportion. Also advises the patient to consume at least 40 units of vitamin A and 50 units of vitamin Daily. The cost of tonics X and Y and the proportion of vitamin A and D that present in X and Y are given in the table below. Formulate L.P.P. to minimize the cost of tonics.

Vitamins	Tonics		Daily requirement in units
	X	Y	
A	2	4	40
D	3	2	50
Cost in Rs. Per unit	5	3	

6. An electronic company is engaged in the production of two components C_1 and C_2 that are used in radio sets. Each unit of C_1 costs the company Rs 5 in wages and Rs 5 in material, while each of C_2 costs the company Rs 25 in wages and Rs 15 in material. The company sells both products on one-period credit terms, but the company's labor and material expenses must be paid in cash. The selling price of C_1 is Rs 30 per unit and of C_2 it is Rs 70 per unit. Because of the company's strong monopoly in these components, it is assumed that the company can sell, at the prevailing prices, as many units as it produces. The company's production capacity is, however, limited by two considerations. First, at the beginning of period 1, the company has an initial balance of Rs 4,000 (cash plus bank credit plus collections from past credit sales). Second, the company has, in each period, 2,000 hours of machine time and 1,400 hours of assembly

time. The production of each C_1 requires 3 hours of machine time and 2 hours of assembly time, whereas the production of each C_2 requires 2 hours of machine time and 3 hours of assembly time. Formulate this problem as an LP model so as to maximize the total profit to the company.

The data of the problem is summarized as follows:

Resources/constraints	Components		Total availability
	C_1	C_2	
Budget (Rs)	10/unit	40/unit	Rs 4000
Machine time	3hrs/unit	2hrs/unit	2000 hours
Assemble time	2hrs/unit	3hrs/unit	1400 hours
Selling price	Rs 30	Rs 70	
Cost (wages +material) price	Rs 10	Rs 40	

(Please answer the all the question and submit the answer script to the tutorial room on or before 4th of November 2024)